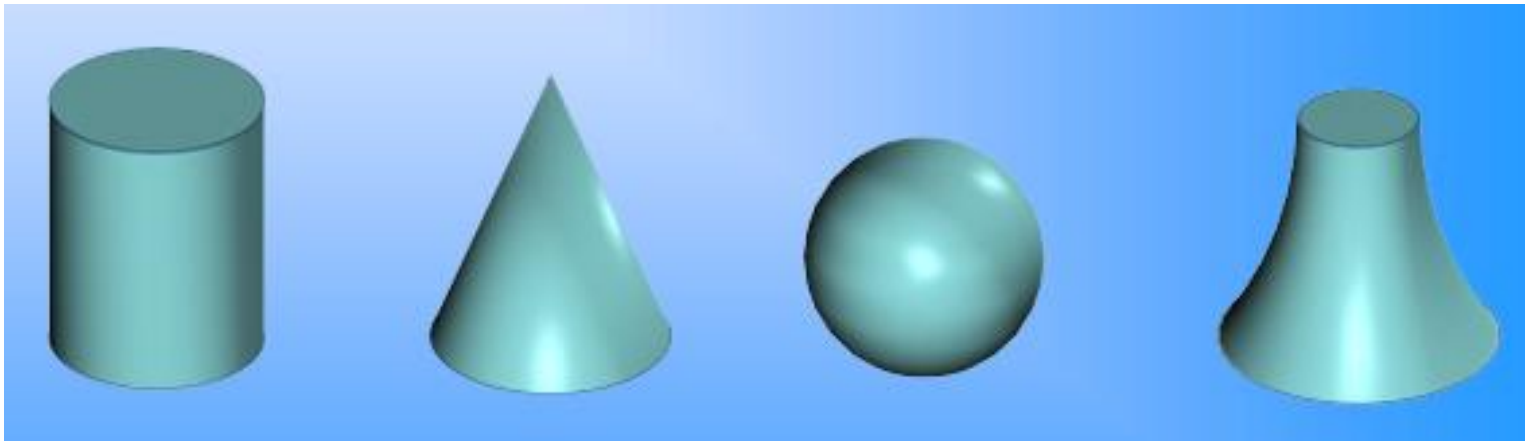
§ 3-3 曲面立体

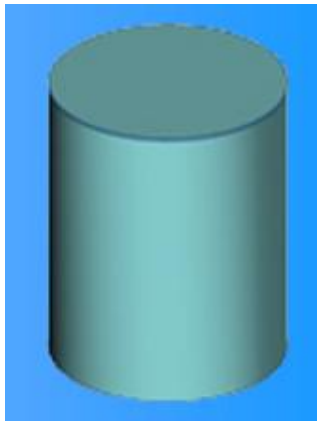
表面由曲面或由平面与曲面围成的立体

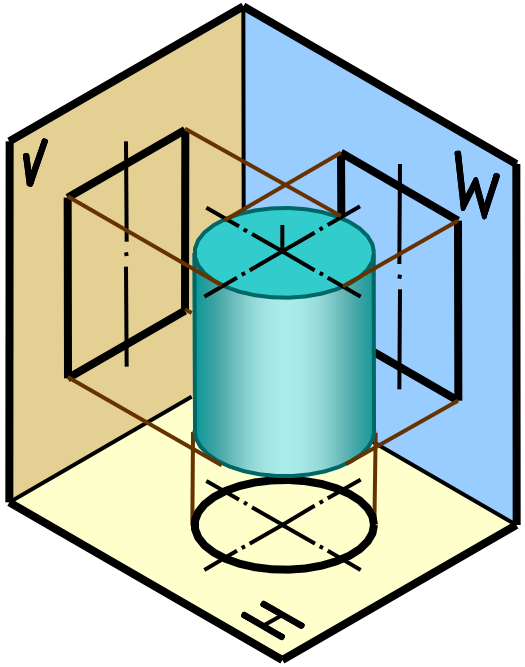


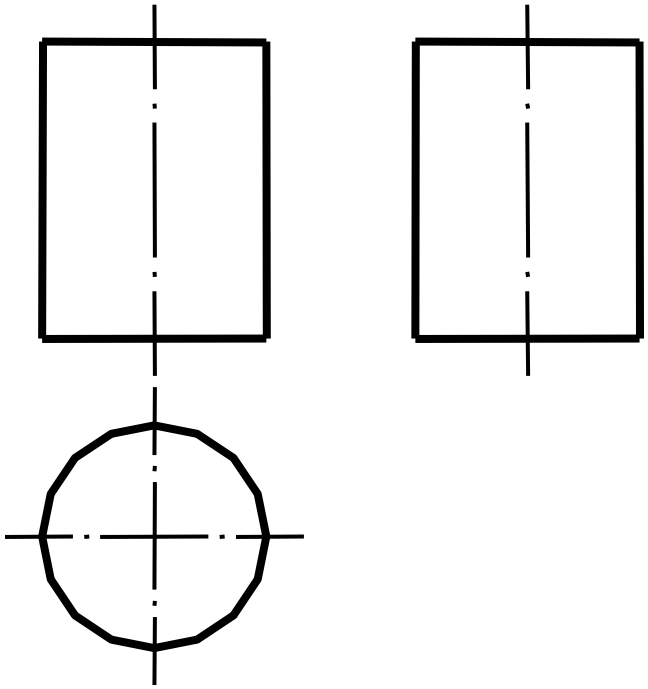
圆柱　　圆锥　　圆球　　圆弧回转体

§ 3-3 曲面立体

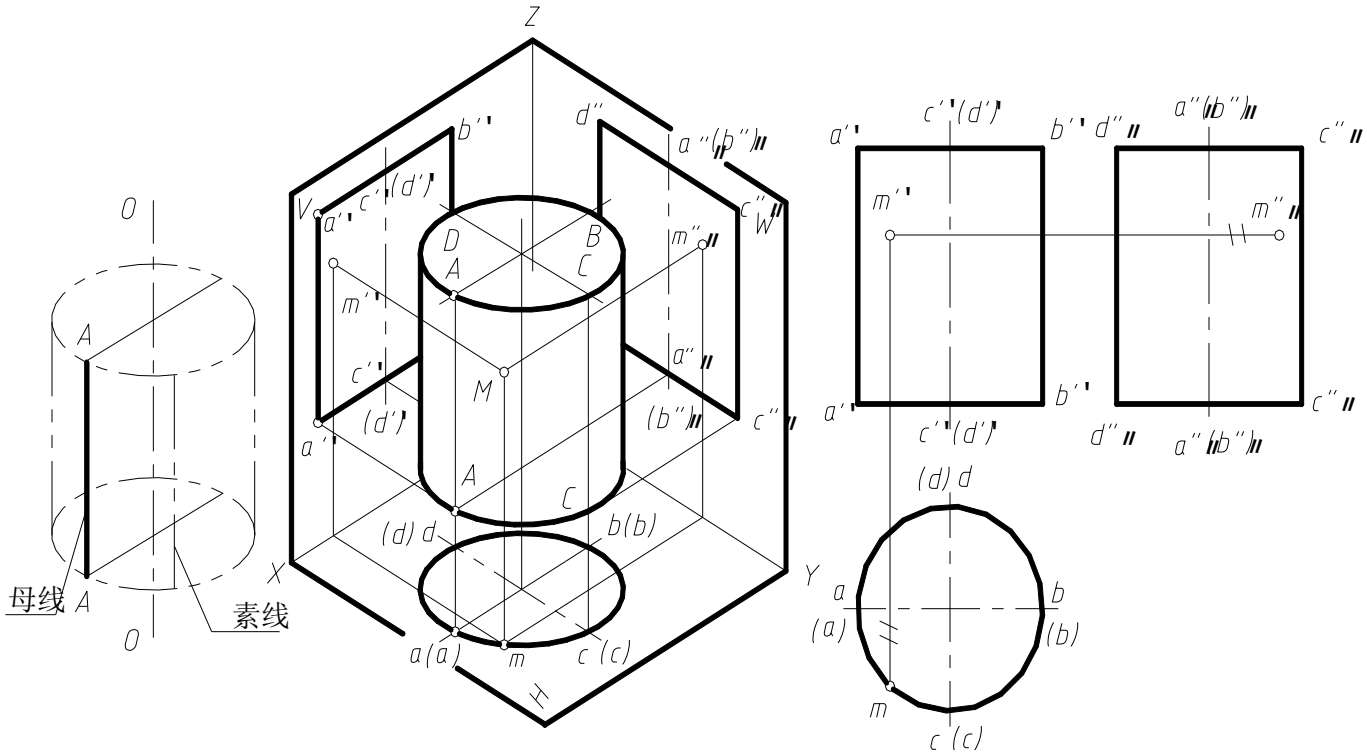
1. 圆柱体的投影及其表面上的点



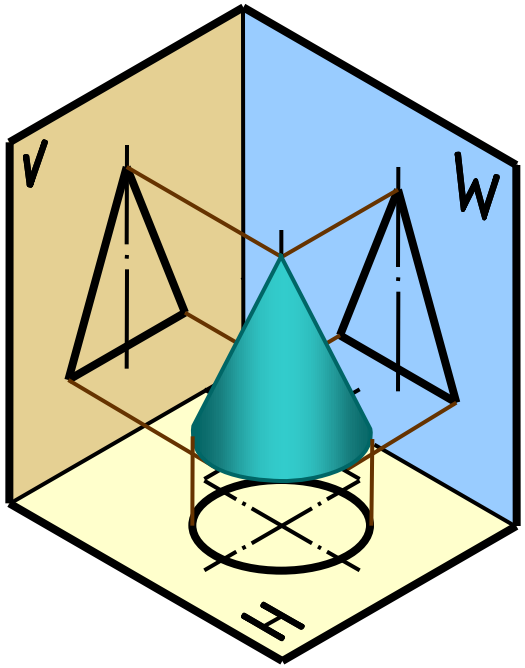


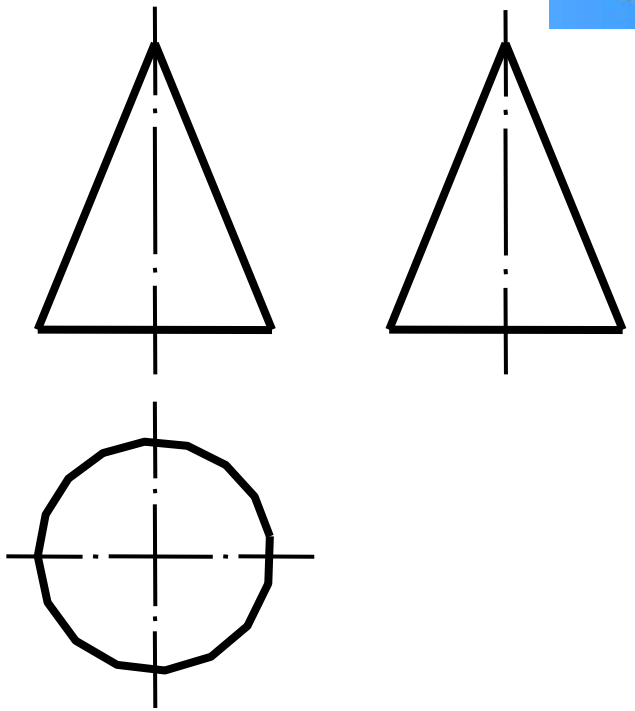


§ 3-3 曲面立体

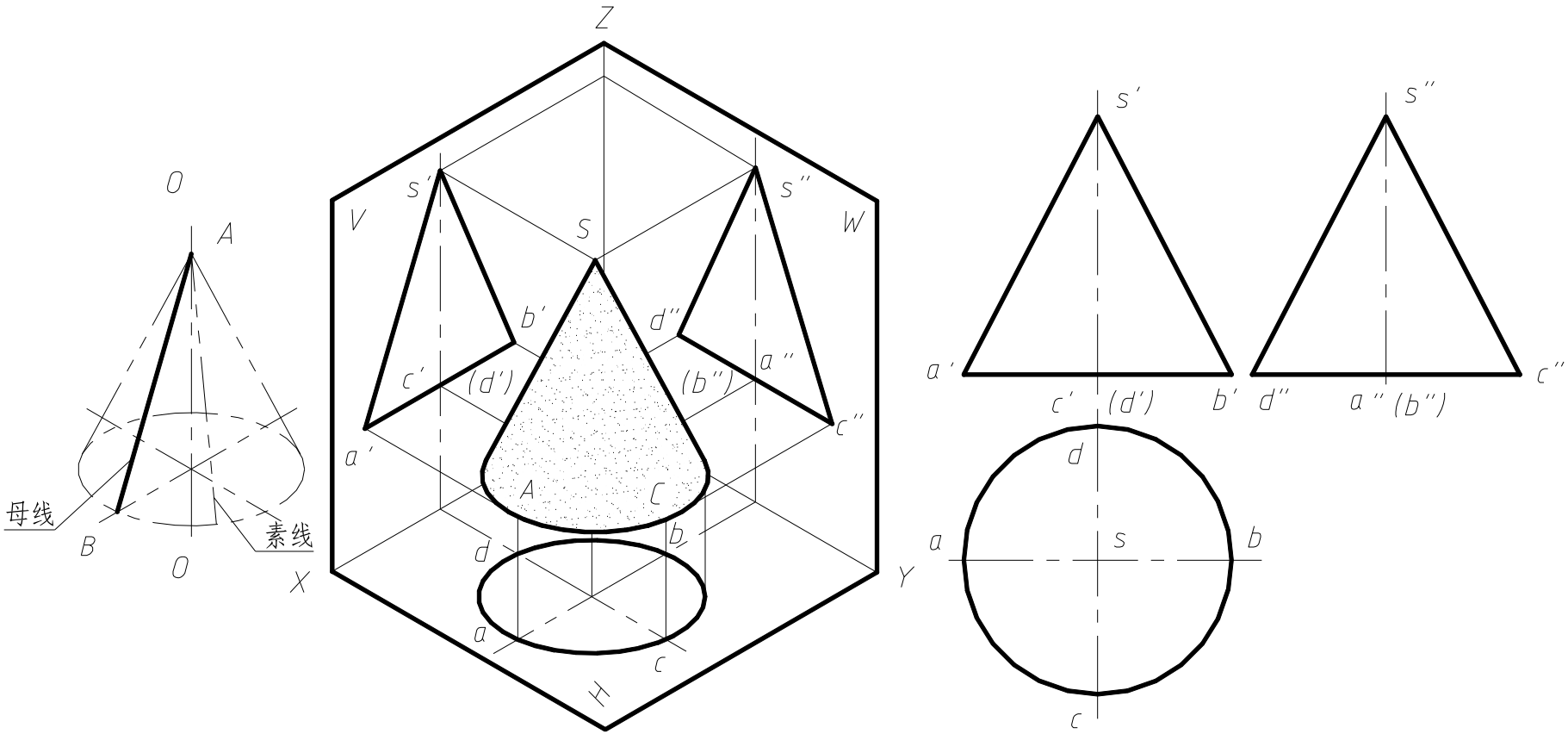


2. 圆锥体的投影及其表面上的点



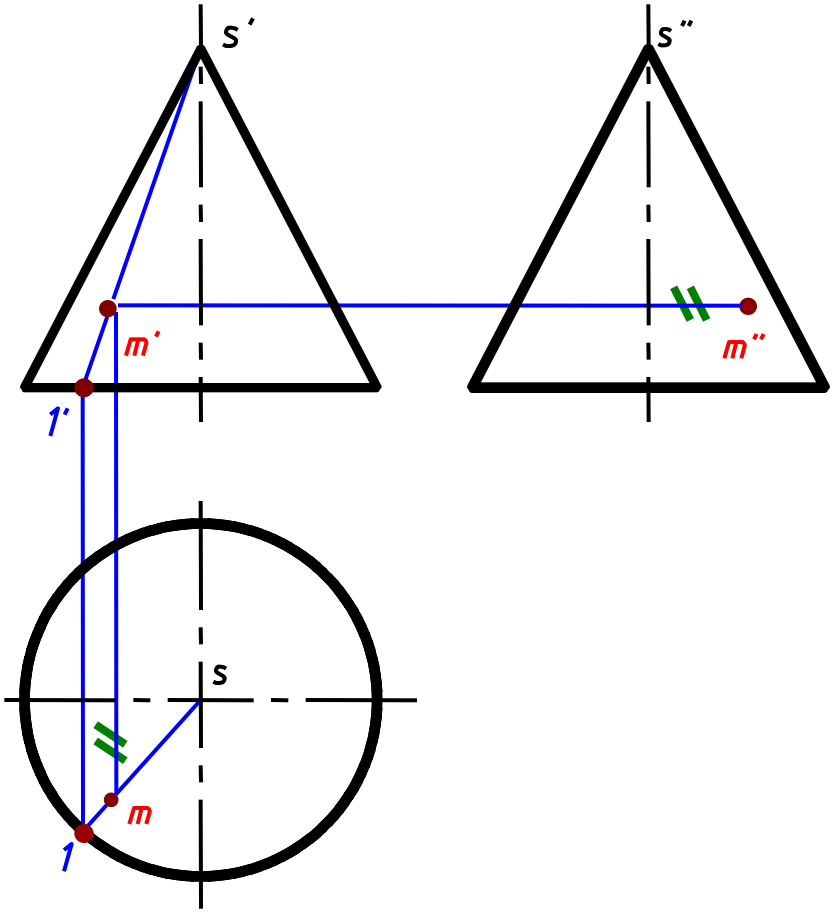


§ 3-3 曲面立体

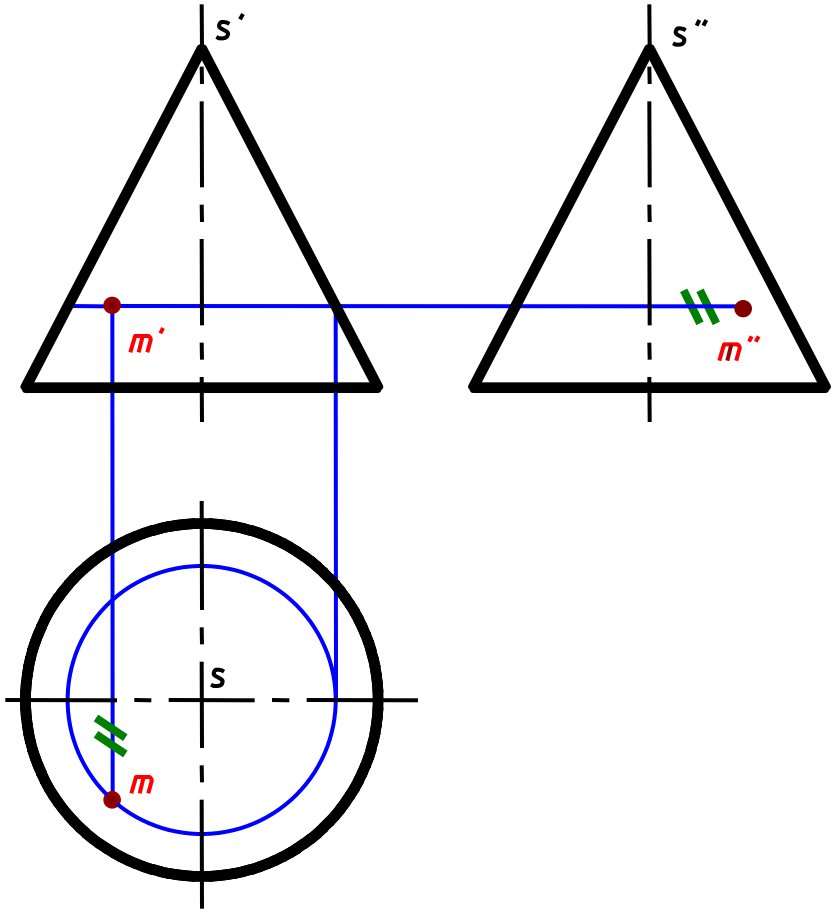


(a) (b) (c)

§ 3-3 曲面立体

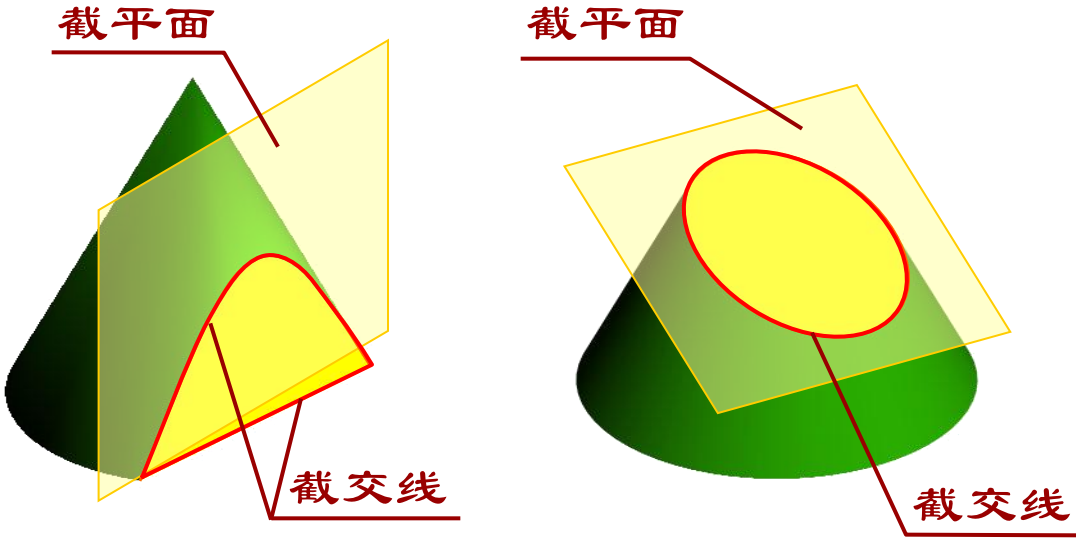


(a)辅助素线法



(b)平行圆法

§ 3-4 平面与曲面立体相交



截平面：截切立体的平面

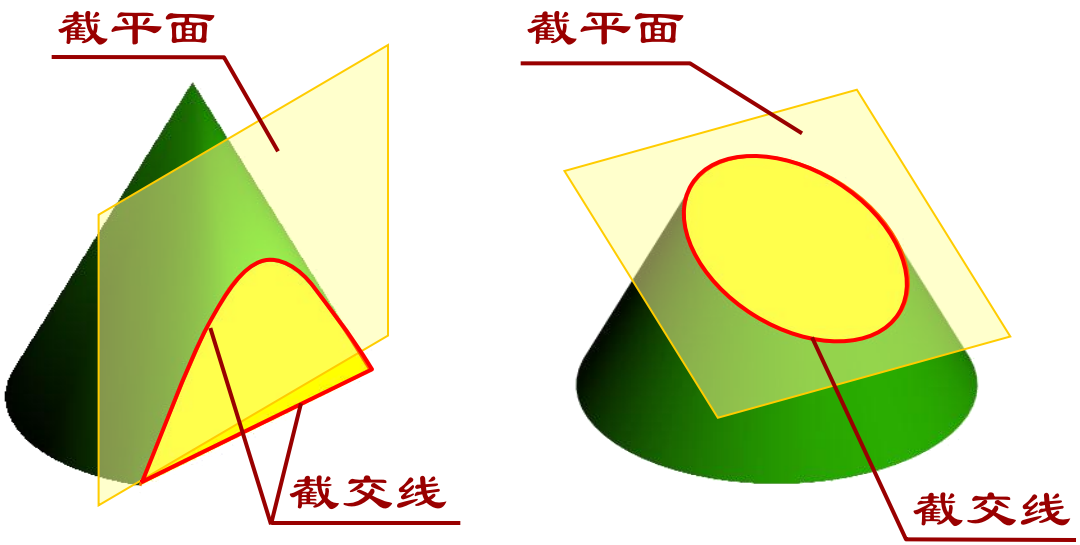
截交线： 截平面与立体表面的交线

求截交线的方法：

1、利用积聚性求截交线。

2、利用辅助平面求截交线。

§3-4 平面与曲面立体相交



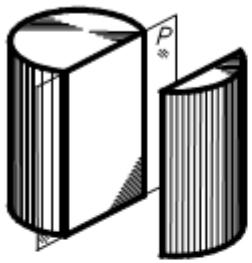
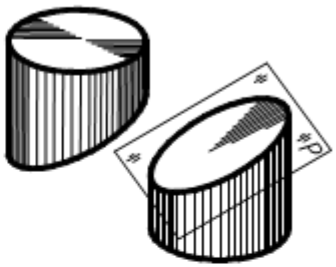
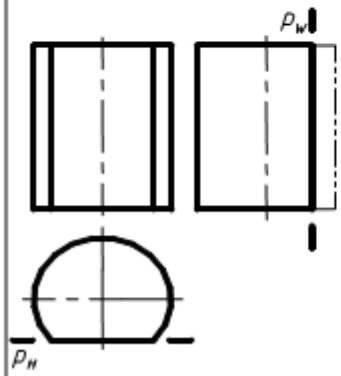
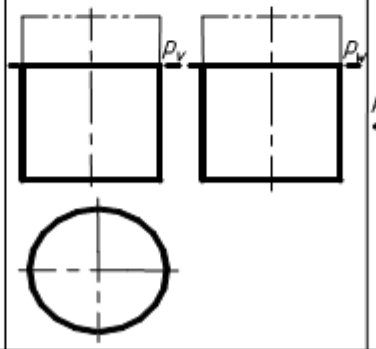
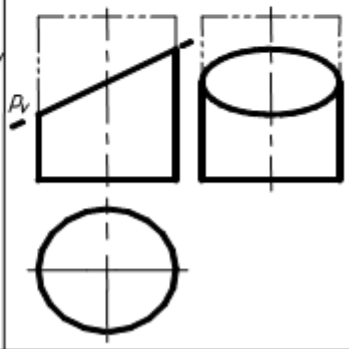
◆ 性质：

共有性 →做题的依据

封闭性 →判断对错的依据

N个截平面,N个封闭图形

平面与圆柱相交

截平面的位置	截平面与圆柱轴线平行	截平面与圆柱轴线垂直	截平面与圆柱轴线倾斜
立体图			
投影图			
截交线	矩形	圆	椭圆

曲面立体截交线的求作步骤：

1. 分析(截平面与立体的相对位置)

2. 作图(→画出截交线的投影) 关键： 找点

椭圆 ➢ 求特殊点

1） 轮廓线上的点 ⟶ 轮廓切否， 截交线可见否

2） 极限位置点 ⟶ 最高低左右前后

3） 截交线本身的特殊点 例：椭圆长短轴

➢ 求一般点 判断走向

➢ 光滑连线， 判别可见性

➢ 整理轮廓线

【例2】圆柱被正垂面所截，补全投影。

作图：

1、找特殊点：

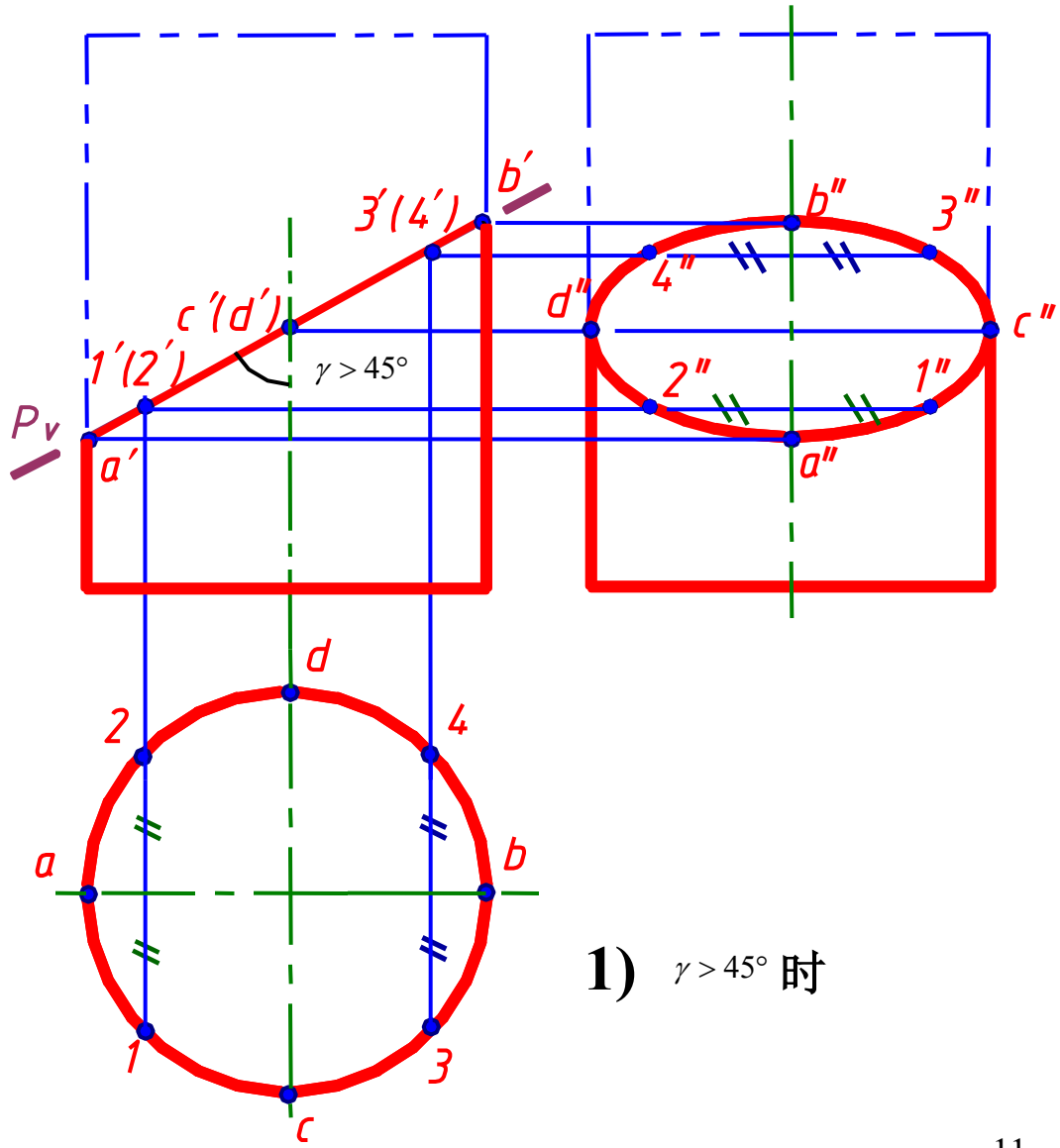
A、B、C、D点

2、取一般点：

1、 2、3、 4点

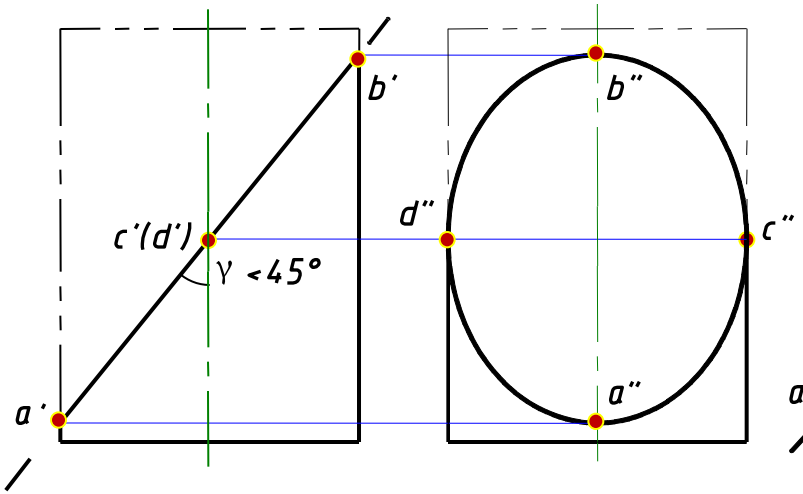
3、光滑连线并判可见性。

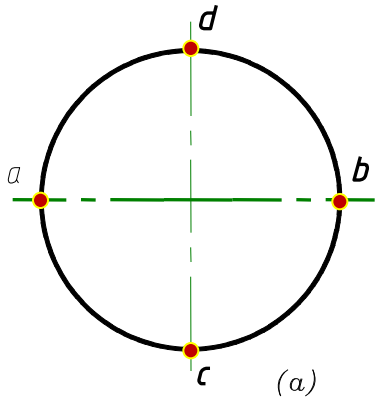
4、整理轮廓线。



1) $\gamma > 45°$ 时

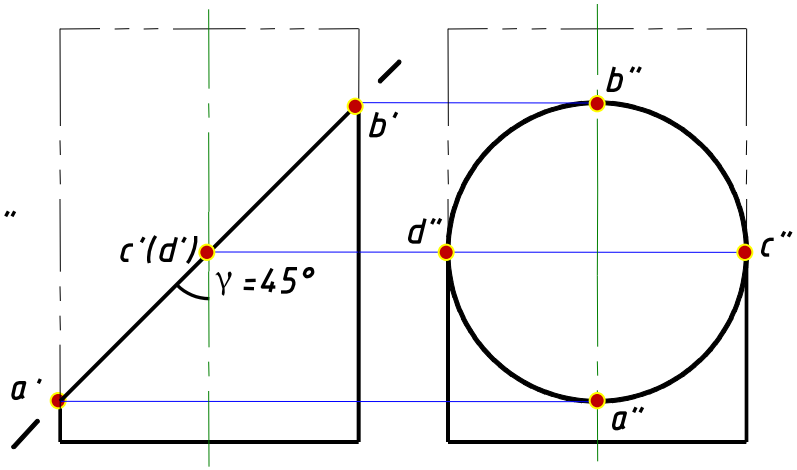
2) $\gamma < 45°$ **时**

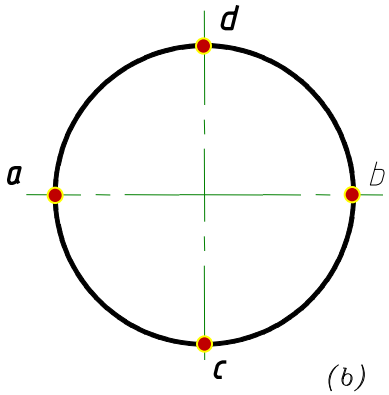




(a)

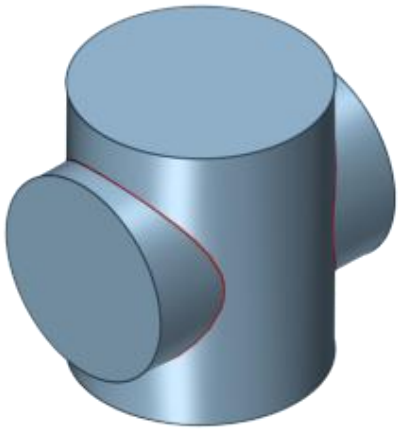
3) $\gamma = 45°$ **时**

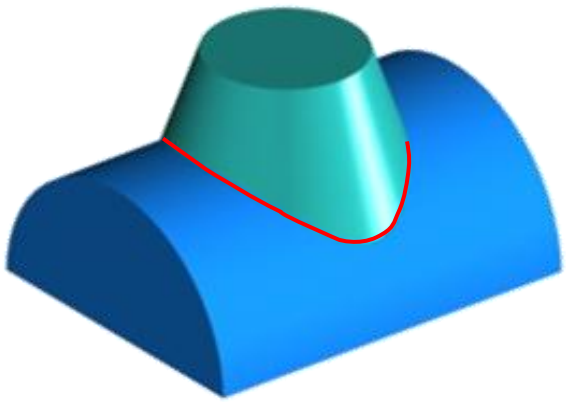




(b)

§3-5 两曲面立体相交



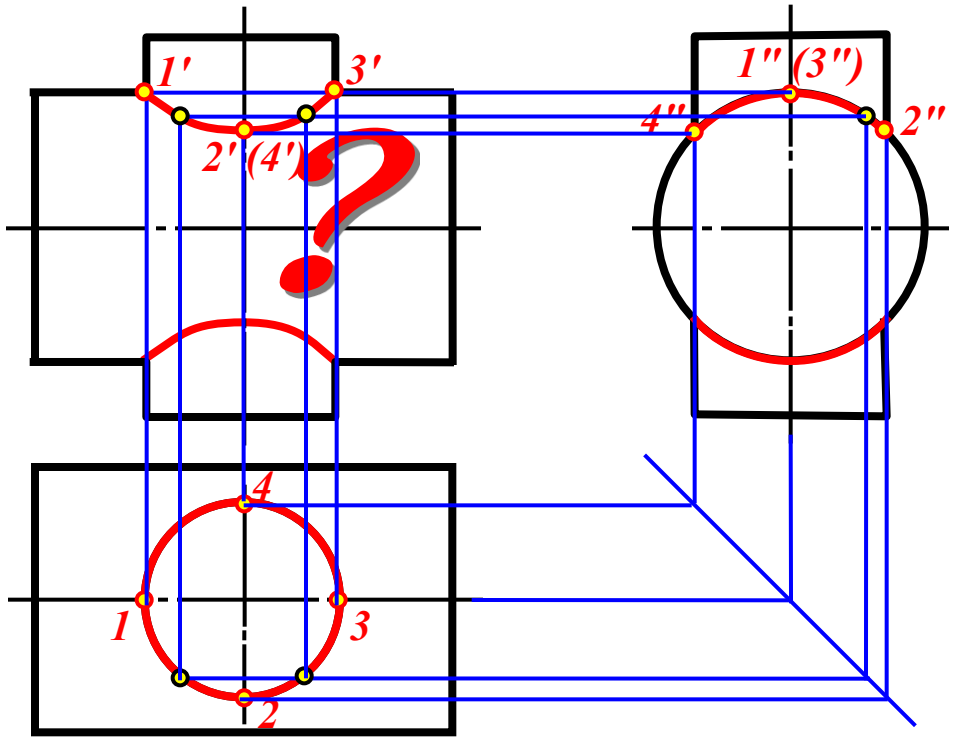


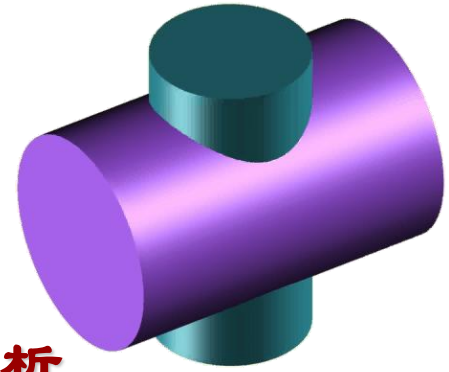
相贯线：曲面立体与曲面立体相交，其表面所产生的交线。

相贯体：相交的曲面立体

相贯线性质：共有性、封闭性

【例4】求正交两圆柱的相贯线





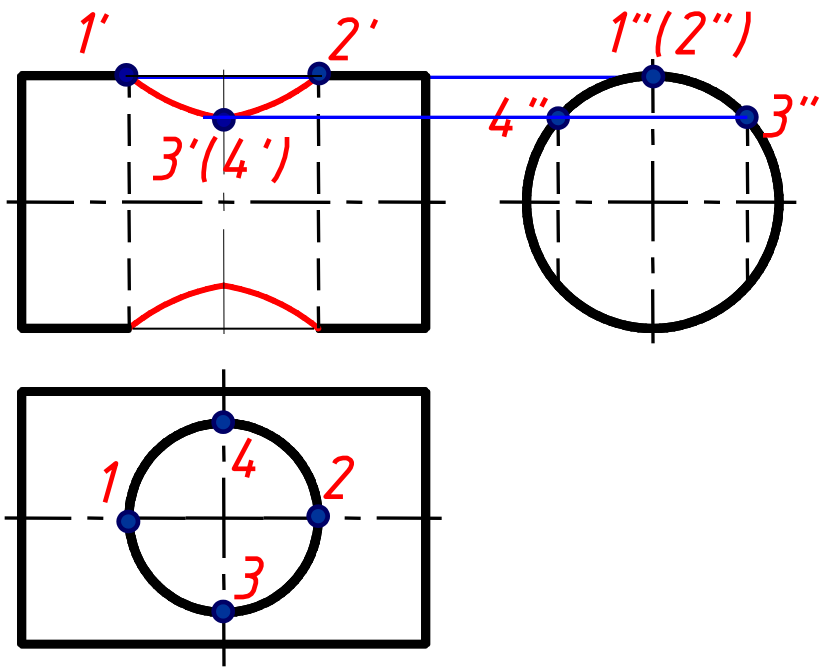
分析

- 空间分析
- 投影分析

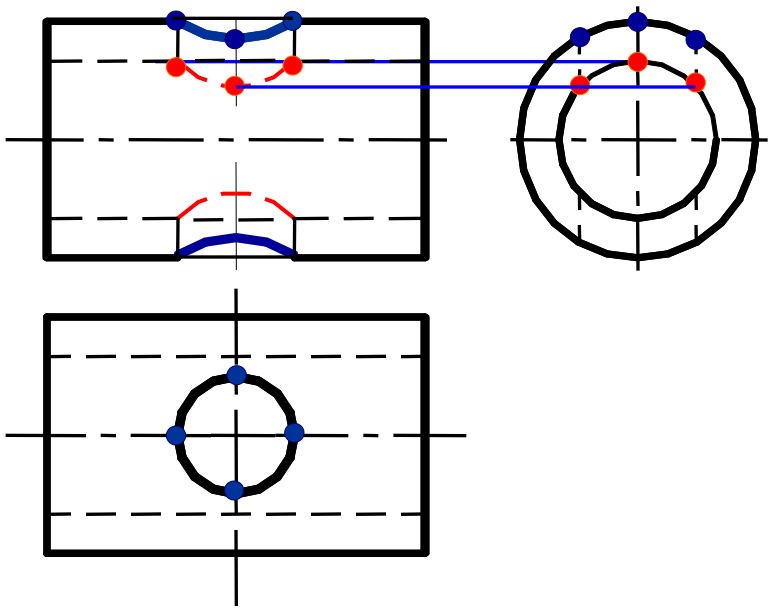
作图

- 作特殊点
  - 转向轮廓线上的点
  - 极限位置点
- 作一般点
- 判别可见性，光滑连接各共有点

【例5】圆柱孔的相贯线。



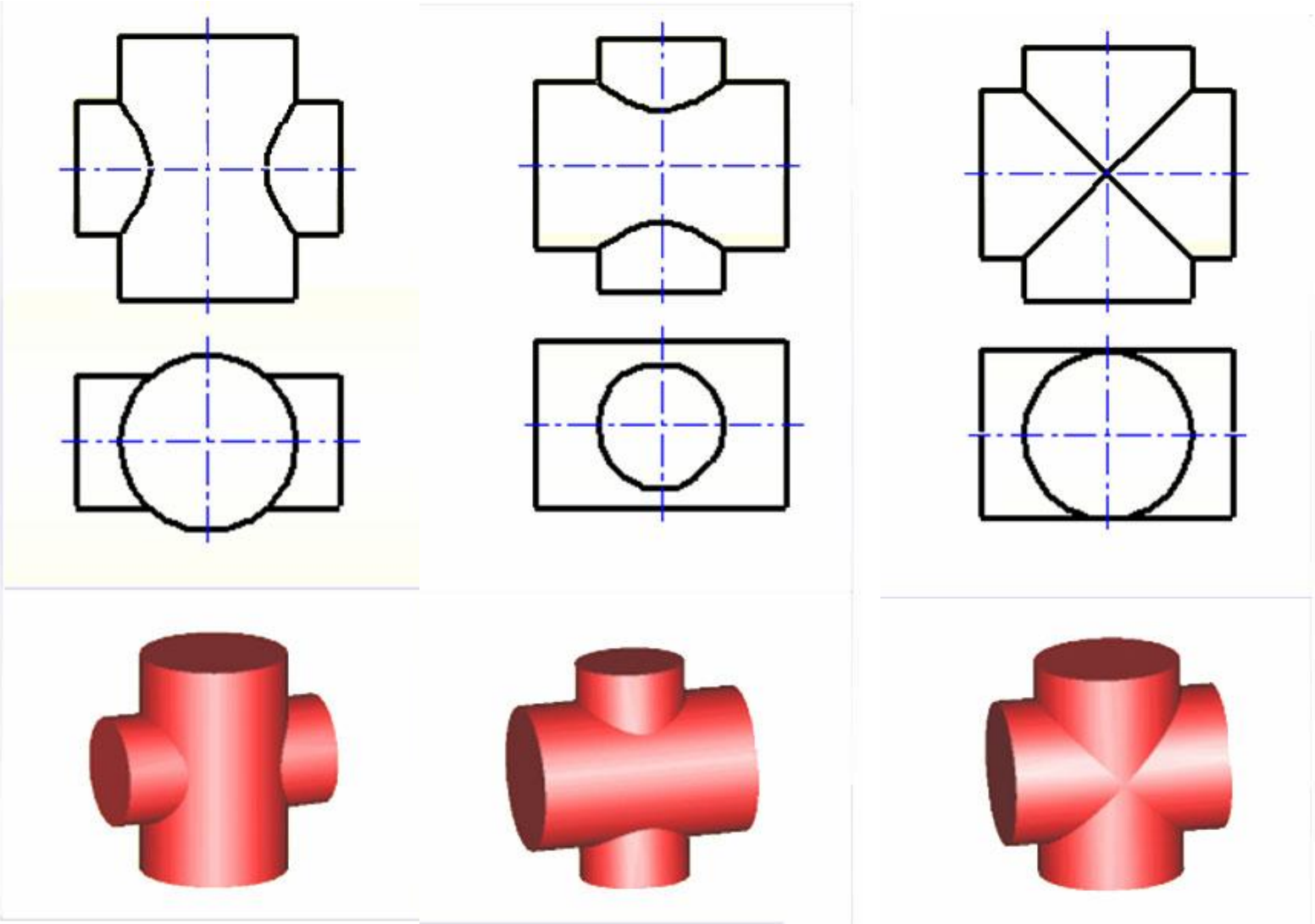
(a)



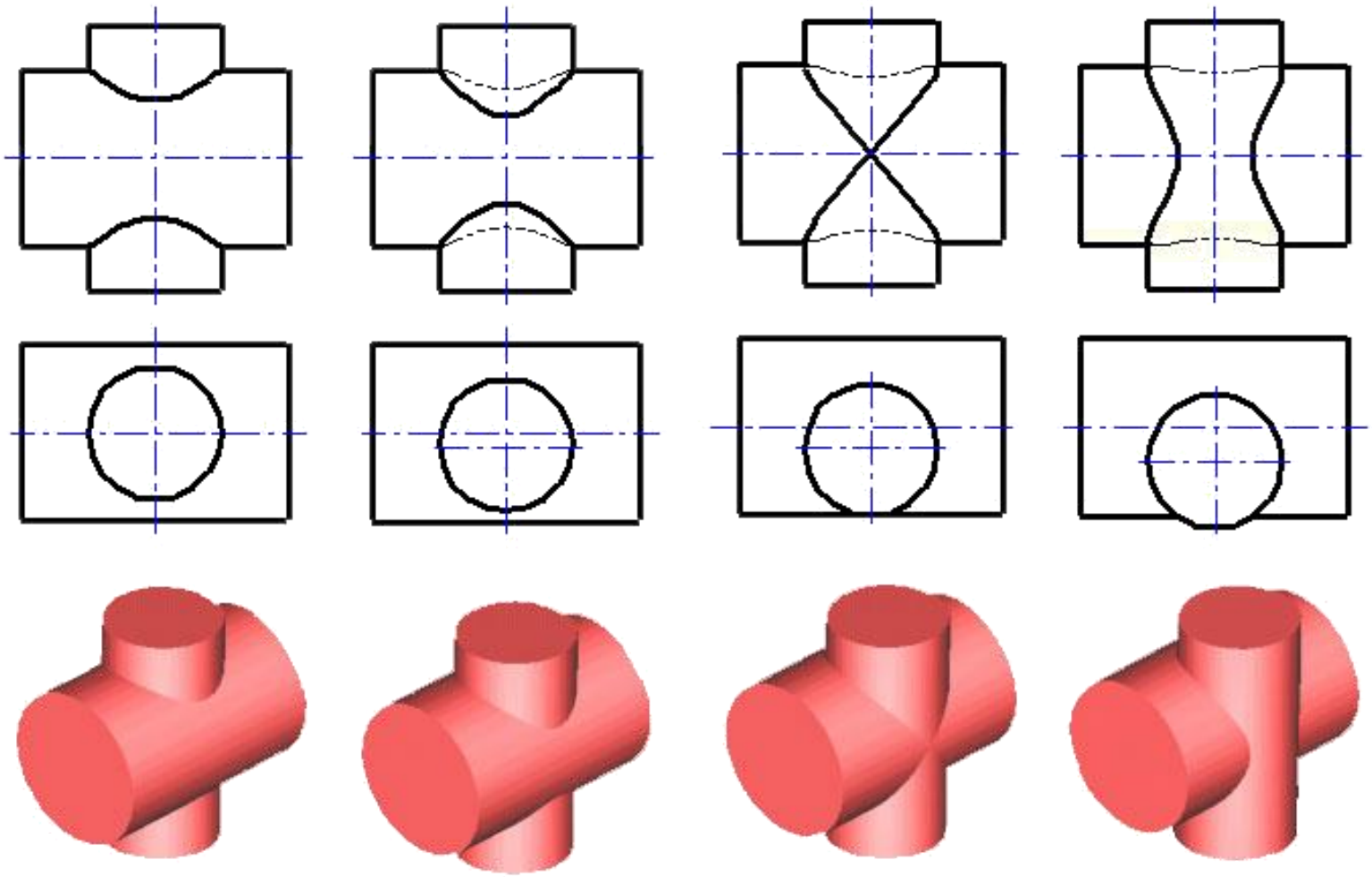
(b)

§3-5 两曲面立体相交

相贯线的变化趋势



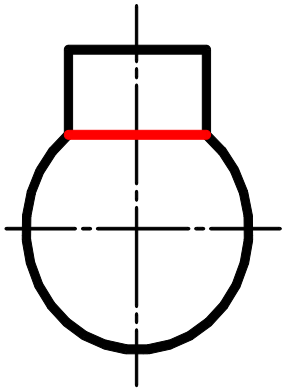
§ 3-5 两曲面立体相交



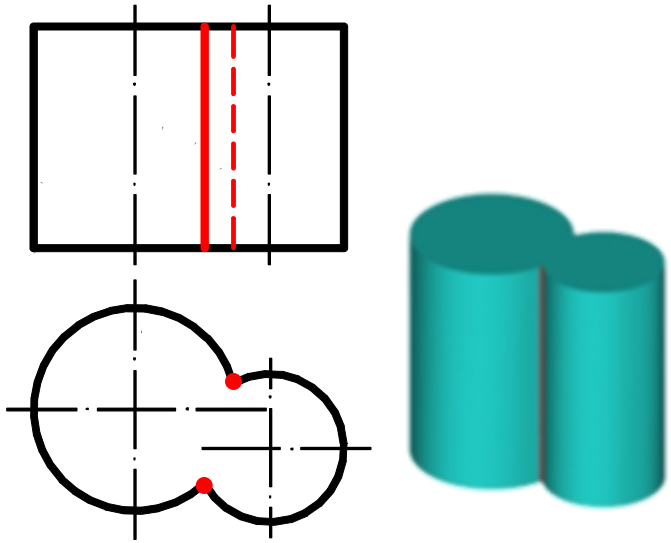
§3-5 两曲面立体相交

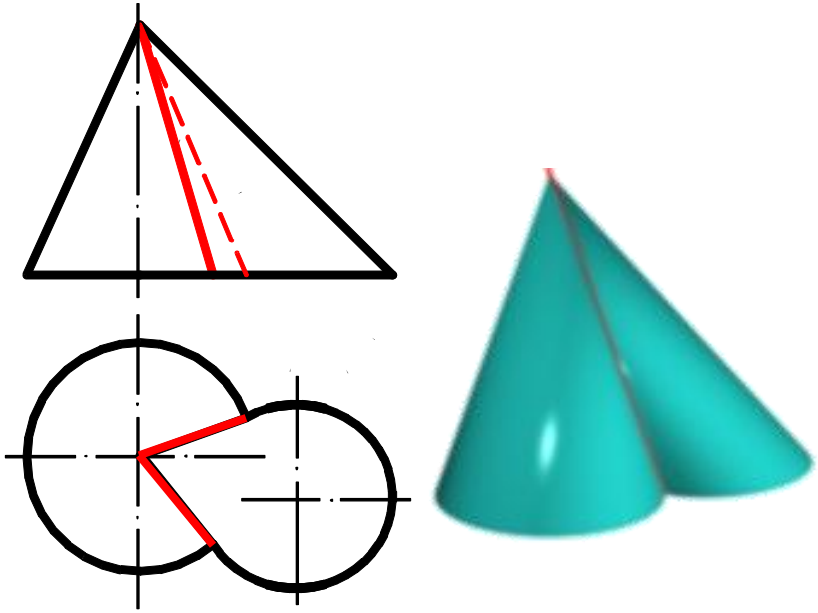
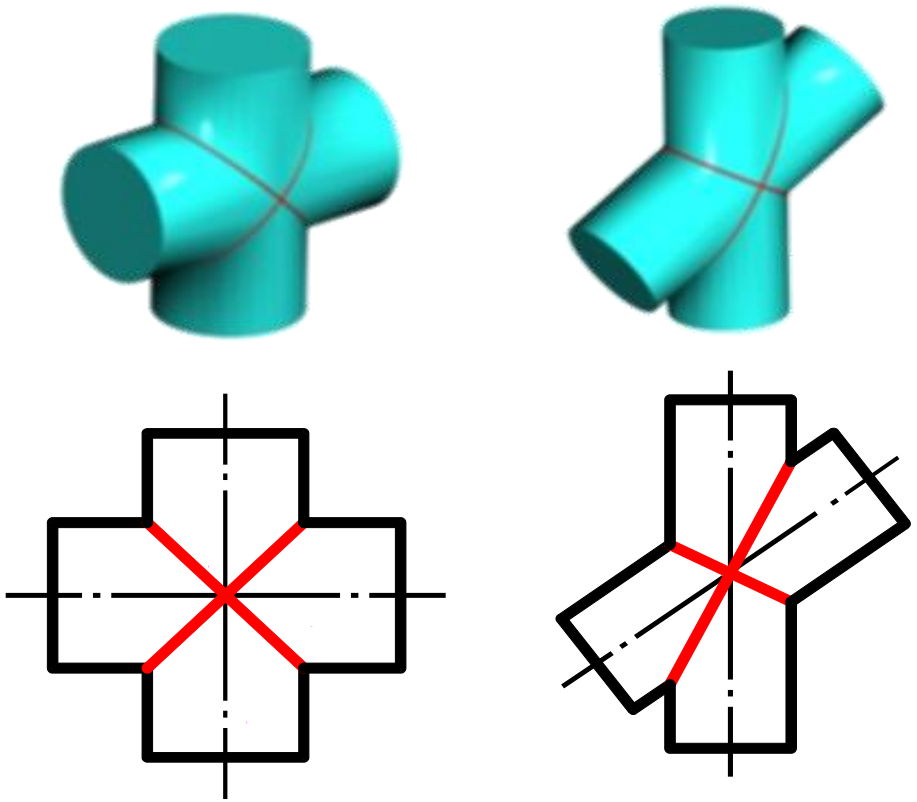
相贯线的特殊情况：

1.同轴相贯，其相贯线为垂直于回转 体轴线的圆。



2. 两圆柱轴线平行或两锥体共锥顶时，其相贯线为直线。





3. 轴线相交的两圆柱，若直径相等（公切于一球面），其相贯线为两条相交的椭圆。